Theia: Team 10

Alex Shaw, Danbi Kim, David Del Val Fraire, Johanna Siebol, Seth Trimberger

Scenario 1: classroom directions

AS-IS
Stevie needs to reach his next classroom.

He knows a left turn is ahead, but cannot judge where to turn.

He turns too early, hits the wall, and hurts his head.

TO-BE
Stevie selects his classroom.

Theia finds his current location and provides a route.

Spoken guidance tells him, “Walk 10 steps. Then turn left.”

Scenario 2: fall assistance

AS-IS
Mary is going to a coffee shop before class.

She slips on paper and falls.

She must wait for someone to notice that she needs help.

TO-BE
Theia presents a fall alert with a 30-second countdown.

“I'm okay” cancels the alert and timer.

“Call for help” or an unanswered countdown opens Phone. Mary completes the call there.

Scenario 3: known obstacles

AS-IS
Bob is heading to a meeting room near a support pillar.

He cannot judge where to turn to avoid it.

He continues straight, hits the pillar, and drops his belongings.

TO-BE
Bob selects the meeting room.

Theia uses the pillar's recorded position to guide his route.

Near the pillar, it says, “Take 2 steps to the left.”

Scenario 4: voice interaction

AS-IS
Laura uses a guide dog and needs to find a meeting room.

Room information is not available in an accessible format.

She must ask another person for directions.

TO-BE
Laura speaks the room number.

Theia repeats the recognized destination for confirmation.

It provides spoken directions while Laura continues using her guide dog.

Priority scenario: emergency assistance

Why this priority?
Delayed help can leave an injured user without assistance.

Fast access must preserve the user's control.

What if?
False alarm: cancel with “I'm okay”.
Unclear speech: retry or use touch.
No response: open Phone after 30 seconds.
No connection: do not claim success.

Who?
Theia presents the alert and timer.
The user requests or dismisses help.
The Phone app controls the call.

What kinds of?
A possible fall.
Another emergency: manual help request.
A false alarm.

When and how assistance proceeds

When?
A possible fall starts a 30-second response window. Manual help is available on request.

How?
“I'm okay” cancels the alert and timer.
“Call for help” opens the Phone app. The user completes the call.
At timeout, Phone opens automatically.

Resulting requirements
Accessible voice and touch controls, cancellation, and clear handoff feedback.
Fall detection needs validation. A calling connection is not guaranteed.